AI Reliability as an Optimization Problem

MATHEMATICAL BRIEF

Optimizing the university pipeline: why the grader dictates learning, proctors stop cheating, and Pass@1 is truth.

THE EDUCATION PIPELINE: RE-RANKING COMPONENT LEVERAGE

Textbook / Prompt	Office Hours	The Proctor	The Grader
CONTEXT & SYLLABUS	LINTERS & COMPILERS	SANDBOX ISOLATION	DETERMINISTIC INVARIANTS
Leverage: Low	Leverage: High	Critical: Anti-Cheat	Leverage: Maximum

FORMAL OPTIMIZATION FORMULATION

CONSTRAINED OBJECTIVE

OBJECTIVE FUNCTION

$$\max_{\theta} \; \mathrm{E}_{x \sim D}\left[Pass@1(x) \cdot I_{\text{clean}} \right] - \lambda \cdot \mathrm{Var}_{\text{run}} - \gamma \cdot \text{Cost}$$

$\partial \mathrm{T}_{\text{test}} / \partial \text{Agent} = 0$ **Proctor Invariant:** Test harness is read-only (zero test-editing).

$\mathrm{P}(\text{Silent Failure}) = 0$ **Fail-Safe Invariant:** Unsolvable problems trigger deterministic halts.

$\mathrm{D}_{\text{unfair}} = \emptyset$ **Fair Exam Filter:** Ambiguous or ill-posed questions purged from test set.

DECISION VARIABLES $\Theta = \{ G, P, F, C \}$

- G **Grader:** Property-based invariants & contract checks (highest leverage).
- P **Proctor:** OS sandbox isolation & write-protection on test files.
- F **Feedback:** Office hours error-contrast loops (compiler & linter turns).
- C **Curriculum:** Task diversity & domain grounding (textbook/prompt).

The Grader > The Teacher

Lectures fail silently. Agents do not optimize against prompt prose; they optimize strictly against the **grader's loss landscape**. A strict compiler beats a 1,000-page syllabus every time.

G has higher gradient than C

Cheating by Exhaustion

Final scores saturate via brute-force retries or test rewriting. **Pass@1 (First Try)** is the only unpolluted signal. Giving agents infinite attempts just rewards random drift.

Optimize Pass@1, not Pass@N

The 100% Fallacy

100% on a fixed test bank is **memorization**, not reliability. In open systems, the real goal is *bounded failure severity*: zero silent errors, deterministic fail-safe recovery.

Bound Failure Severity

Lead Principle: "The Proctor Precedes the Professor." Without sandbox guards, the fastest path to 100% is editing the exam.

KEY AXIOM

HYPOTHESIS **Test-Time Compute may just be "All-Night Cramming":** More reasoning tokens boost closed-book scores, but we speculate they increase brittleness against out-of-distribution real-world drift.